Aayojan

Wedding CRM & Operations Platform

Plan. Organize. Celebrate.

Document Type	Technical Report
Date	March 21, 2026
Platform	Web Application
Status	All features implemented and functional

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1. Executive Summary

Aayojan is a full-stack web application purpose-built for professional wedding planners in the Indian market. The platform consolidates the entire lifecycle of a wedding planning business — from first client inquiry to post-event close-out — into a single, role-aware interface.

The system replaces the patchwork of WhatsApp threads, spreadsheet trackers, and disconnected email chains that currently define the industry's operational baseline. It delivers structured lead management, wedding event coordination, vendor communication automation, budget tracking, and AI-assisted planning in one cohesive product.

The technical foundation is a modern, production-grade stack: React 19 with Zustand on the frontend; Express and MongoDB on the backend; with optional integrations for Groq (AI), Twilio (SMS), Nodemailer (email), TripAdvisor via RapidAPI (hotels), and Pixabay (mood board imagery). All core CRM features function without any third-party integration.

2. Problem Statement

Wedding planning in India is a high-stakes, high-complexity service business. A single wedding involves dozens of vendors, multiple sub-events (Mehendi, Sangeet, Baraat, Reception), hundreds of tasks, and a client relationship spanning six to eighteen months. Planners managing three to ten simultaneous weddings face three compounding operational problems:

1. **Scattered communication.** Client conversations happen over WhatsApp, vendor confirmations over phone calls, and task assignments over informal messages. There is no single source of truth.
2. **No visibility into progress.** Without a centralized tracker, managers cannot determine at a glance which tasks are overdue, which vendors are unconfirmed, or which weddings are behind schedule.
3. **Operational chaos under scale.** As a planning firm grows, the absence of role-based access and structured delegation creates duplication of effort, missed deadlines, and client-facing errors.

Existing general-purpose CRM tools — Salesforce, HubSpot, Zoho — are not designed for event-centric workflows and require significant customization to approximate what a wedding planner actually needs. No purpose-built, professionally designed solution existed for this segment before Aayojan.

3. Solution Overview

Aayojan addresses these problems through a unified operations platform built on four design principles:

- **One platform, full lifecycle.** Every action, document, and communication is logged and accessible in one place, from first inquiry through event close-out.

- **Structured role separation.** Four distinct roles — Admin, Relationship Manager, Team Member, and Client — each see only what they need and can act only within their authority.
- **Automation at critical junctures.** Vendor reminders, task-due notifications, and progress nudges are automated via scheduled cron jobs, email, and SMS, reducing the manual overhead of follow-up.
- **Template-driven speed.** One-click wedding templates (Destination, Local, Luxury, Intimate) pre-populate events and tasks, cutting setup from hours to seconds.

4. System Features

4.1 Lead Pipeline Management

The lead module implements a Kanban-style pipeline across four stages: Inquiry, Proposal, Negotiation, and Booked. Leads store contact details, budget, source channel, assigned manager, and a full activity log. A drag-and-drop interface (via `@dnd-kit`) allows stage updates without navigating away from the board. Pipeline analytics — total value, conversion rate, stage funnel, and source distribution — are rendered inline using Recharts.

4.2 Wedding & Event Management

Each confirmed booking converts to a Wedding record containing client metadata, date, team assignment, and a linked lead. Within a wedding, planners create Events (sub-occasions such as Mehendi or Reception), each with its own task list, vendor assignments, document attachments, and linked hotel accommodations. A progress percentage is computed server-side from task completion ratios and surfaced throughout the interface.

4.3 Task System with Maker-Checker Workflow

Tasks are the operational unit of Aayojan. Each task carries a category, priority level, due date, assignee, linked vendor, subtask list, and document attachments. A two-step completion flow — mark-done by an assignee, then verify by a manager — ensures accountability without bureaucratic friction. Overdue and high-priority tasks are surfaced prominently in the dashboard and notification feed.

4.4 Vendor Directory

A searchable, category-filtered vendor database stores contact details, city, pricing tier, ratings, and linked tasks. Vendors are reusable across weddings. The notification scheduler dispatches automated email (Nodemailer) and SMS (Twilio) reminders to vendors for upcoming task deadlines, running every 30 minutes via `node-cron`.

4.5 Budget Tracker

A per-wedding financial overview aggregates vendor payments, displays pending versus completed amounts, and tracks payment status at the task level. The module provides a clear financial snapshot without requiring external spreadsheet tooling.

4.6 Hotel Search & Event Linking

Powered by the TripAdvisor API via RapidAPI, the hotel module allows planners to search accommodations by city and date, view ratings and pricing, and link results directly to a wedding event, eliminating context-switching to external booking platforms.

4.7 Mood Board

A Pinterest-style inspiration board allows planners and clients to collect visual references. Images can be uploaded directly or sourced via Pixabay. Boards are linked to weddings and optionally to specific events, giving clients a visual artifact of the planning process.

4.8 Wedding Templates

Four pre-built templates (Destination, Local, Luxury, Intimate) each define a complete wedding structure: events, task categories, and checklists. One-click conversion instantiates a fully scaffolded wedding record, turning setup from a blank-slate exercise into a refinement task.

4.9 AI Planning Assistant

An in-app chat assistant powered by Groq's `llama-3.1-8b-instant` model receives live context about the user's weddings, tasks, and leads on each query. It answers operational questions, suggests vendor categories, and drafts communication. A polling hook (`useWeddingPoller`) surfaces proactive nudges when high-priority items require attention.

4.10 Role-Based Access Control

Four roles with granular permissions govern every action. Admins have unrestricted access. Relationship Managers handle assigned leads and weddings. Team Members act on assigned tasks. Clients view progress, mark tasks complete, and upload documents. JWT tokens carry role claims; API middleware enforces permissions at every endpoint.

5. System Architecture

The application follows a three-tier architecture: a React single-page application communicates over REST with an Express API server backed by MongoDB.

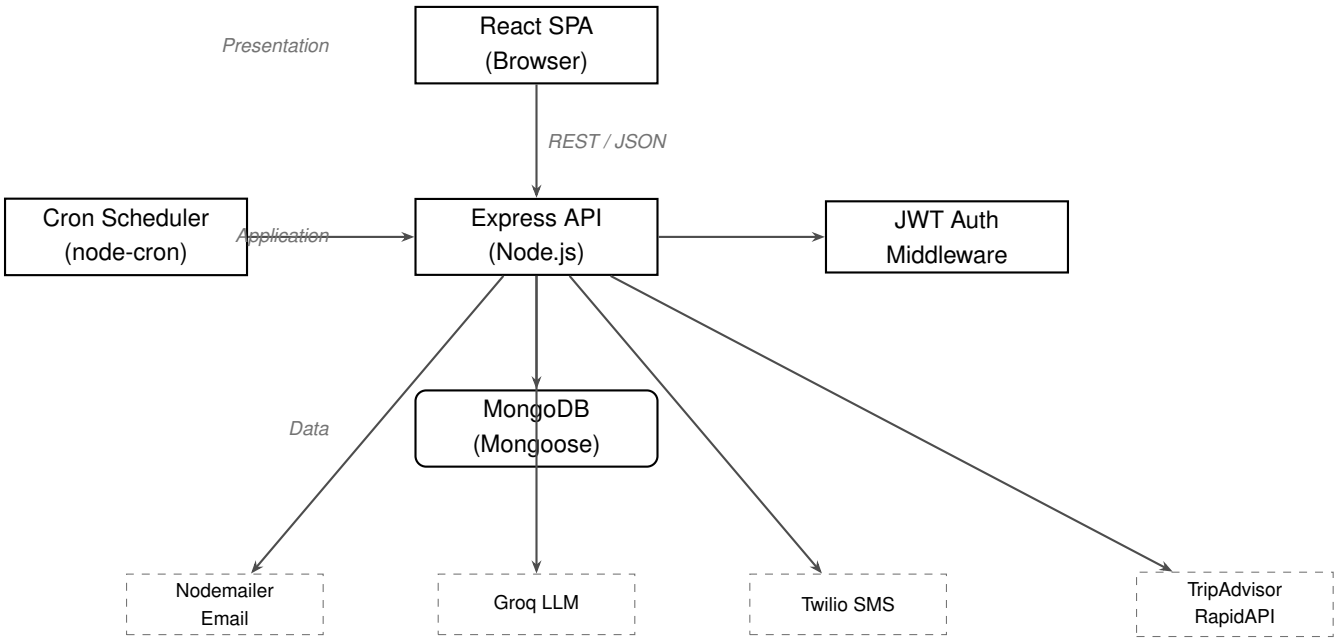

Figure 1. *High-level system architecture of Aayojan*

5.1 Frontend

The React frontend uses Zustand for global state, organized into domain-specific stores (auth, wedding, lead, task, vendor, budget, notifications). React Router v7 handles client-side routing with protected route wrappers enforcing role-based access at the navigation layer. Axios is configured with JWT interceptors that attach the bearer token to every request and redirect to login on 401 responses.

5.2 Backend

The Express server exposes a RESTful API organized by resource domain. Mongoose schemas enforce data integrity and define relationships between Users, Weddings, Events, Tasks, Leads, Vendors, and Notifications. A `notificationScheduler` module runs as a background cron job, querying for tasks approaching their deadlines and dispatching email and SMS alerts to assigned vendors.

5.3 Database Design

MongoDB's document model accommodates the hierarchical, variable-structure data of wedding planning naturally. Weddings embed event references; tasks reference vendors and users by ObjectId; notifications are user-scoped. The schema supports nested subtasks without join-heavy queries.

6. Tech Stack

6.1 Frontend

Technology	Version	Role
React	19	UI framework; hooks-based component architecture
Vite	8	Dev server and production build tooling
TailwindCSS	4	Utility-first styling; custom warm/stone palette
Zustand	—	Lightweight global state management
React Router	v7	Client-side routing with protected routes
@dnd-kit	—	Accessible drag-and-drop for the Kanban board
Recharts	—	Composable charts for dashboard analytics
Axios	—	HTTP client with JWT interceptors
Groq SDK	—	LLM inference client for the AI assistant
Lucide React	—	Consistent SVG icon library
date-fns	—	Lightweight date parsing and formatting

Table 1. *Frontend technology stack*

6.2 Backend

Technology	Version	Role
Node.js + Express	4	RESTful API server
MongoDB + Mongoose	8	Document database with schema validation
jsonwebtoken	—	Stateless auth; 7-day token expiry
bcryptjs	—	Password hashing; 12 salt rounds
Multer	—	Multipart file upload handling
node-cron	—	Scheduled vendor notification jobs (every 30 min)
Nodemailer	—	SMTP email notifications to vendors
Twilio	—	SMS alerts for upcoming task deadlines

Table 2. *Backend technology stack*

6.3 External Integrations

Service	Feature
Groq (free tier)	AI assistant with live wedding and task context
TripAdvisor (RapidAPI)	Hotel search and event-level linking
Pixabay	Free image search for mood boards
Gmail SMTP	Vendor task-deadline email reminders
Twilio	Vendor SMS alerts

Table 3. *Optional third-party integrations — core CRM works without any of them*

7. UI/UX Overview

Aayojan's interface follows a dark-header, light-content layout that establishes visual hierarchy without relying on color gimmicks. Navigation is persistent in both a left sidebar and a top bar; context-sensitive action buttons are scoped to the current view. The design system uses a stone-and-warm palette via TailwindCSS 4, with typographic spacing that keeps dense operational data readable.

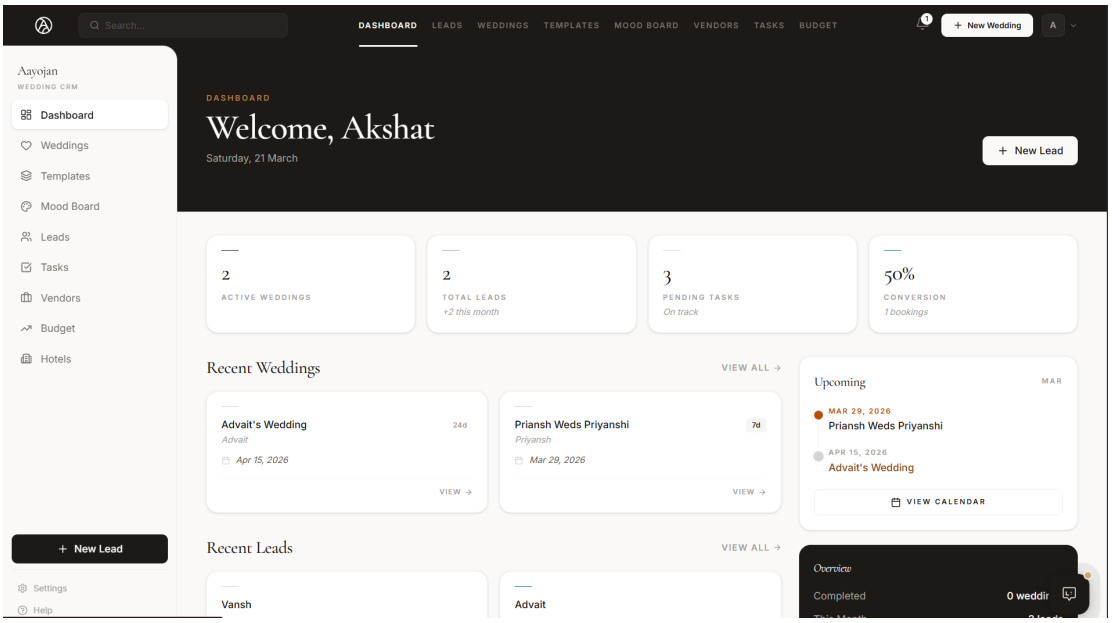

Figure 2. *Dashboard — KPI cards, recent weddings, upcoming event calendar, and lead summary*

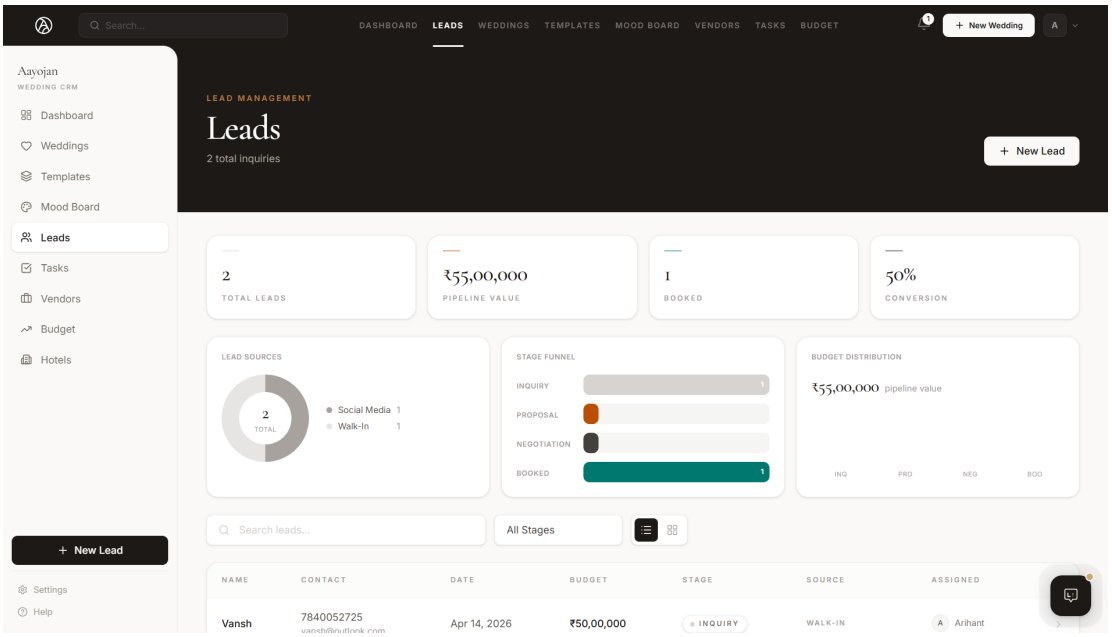

Figure 3. *Lead management — stage funnel, source distribution chart, and sortable lead table*

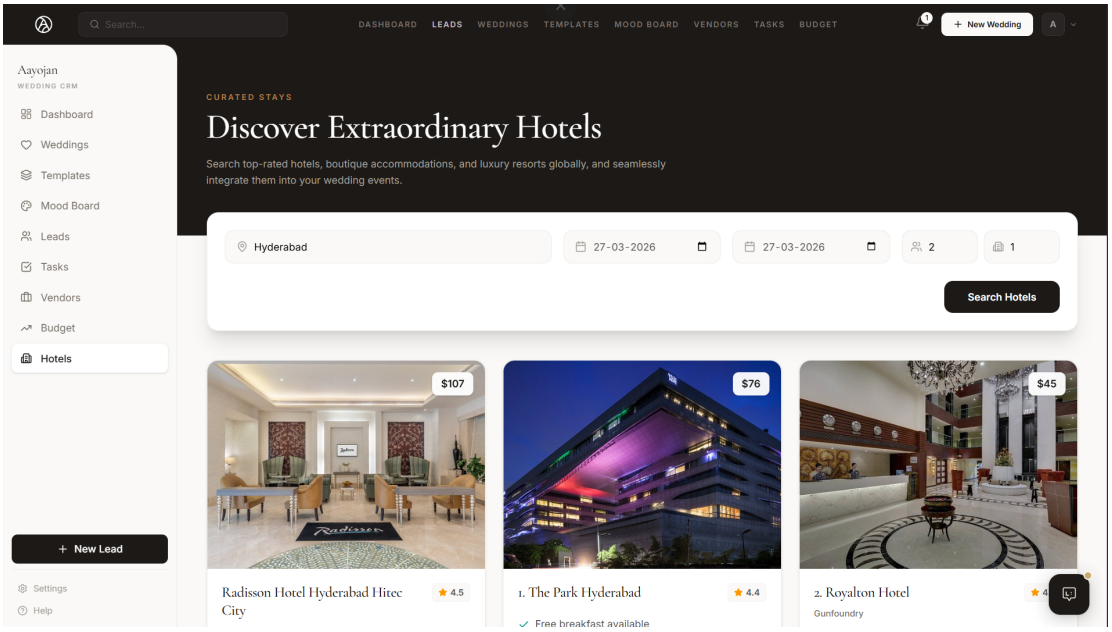

Figure 4. *Hotel search — TripAdvisor-powered discovery with ratings, pricing, and event linking*

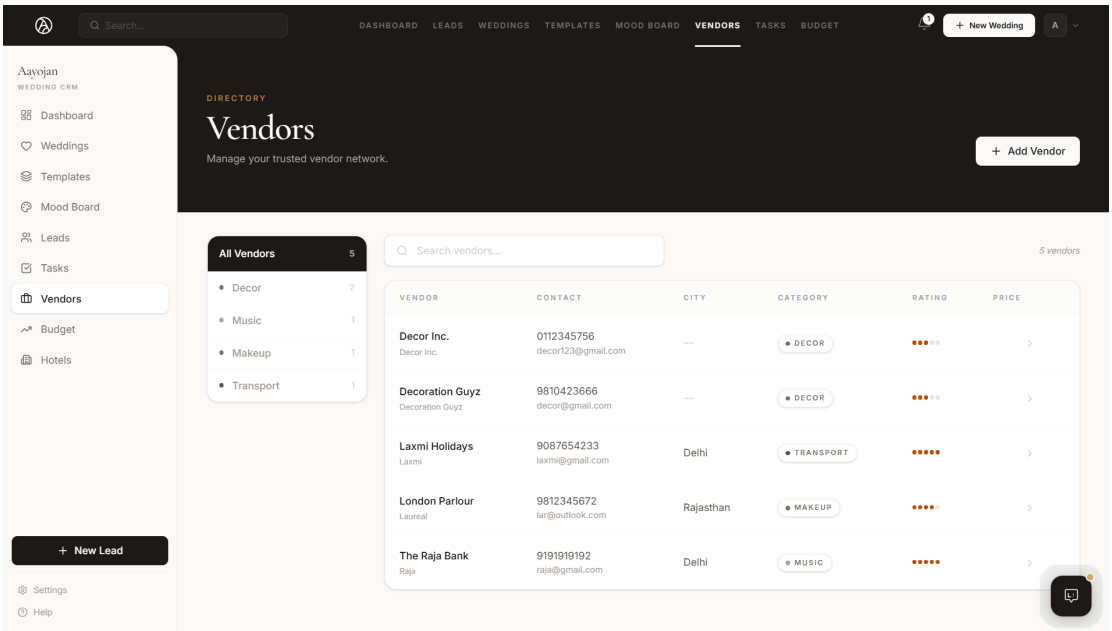

Figure 5. *Vendor directory — category-filtered list with contacts, ratings, and city*

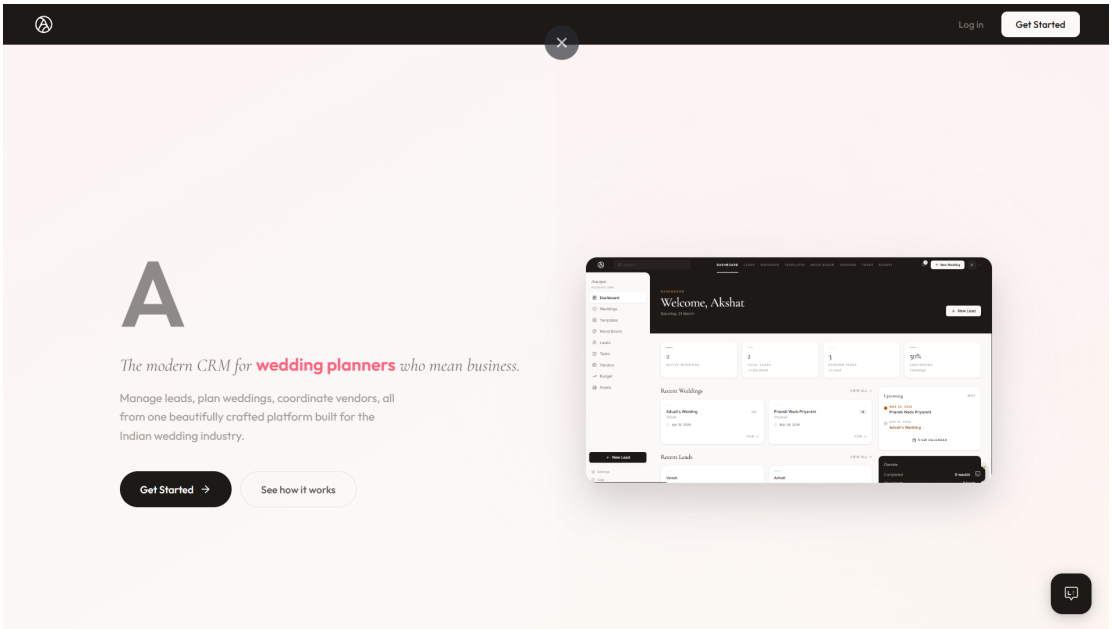

Figure 6. *Landing page — product positioning for professional wedding planners*

8. Key Workflows

8.1 Lead to Wedding Conversion

1. A client inquiry is created as a Lead with source, budget, and contact details.
2. The Relationship Manager moves it through Inquiry → Proposal → Negotiation → Booked via the Kanban board.
3. On booking, a single API call (`POST /api/leads/:id/convert`) instantiates a Wedding record linked to the lead, preserving all context.

4. The wedding is then populated manually or via a one-click template.

8.2 Task Lifecycle with Maker-Checker

1. A manager creates a task, assigns it to a team member or vendor, and sets a due date and priority.
2. The assignee completes the work and marks the task done (`status: done`).
3. A manager reviews and marks it verified (`status: verified`).
4. Only verified tasks count toward the wedding's progress percentage.

8.3 Automated Vendor Notification

1. `node-cron` triggers `notificationScheduler` every 30 minutes.
2. The scheduler queries for tasks with upcoming deadlines that have linked vendors with contact details.
3. Email reminders go out via Nodemailer; SMS alerts via Twilio.
4. An in-app notification record is created alongside each external alert.

9. Innovation

Aayojan's novelty lies not in any single technology but in the deliberate assembly of capabilities around the specific operational model of the Indian wedding planning industry.

Domain-specific data model. The schema hierarchy — Weddings containing Events, Events containing Tasks, Tasks referencing Vendors — mirrors the mental model of a wedding planner, rather than forcing the domain into a generic CRM schema.

Maker-checker at the task level. Applying a two-step verification workflow to individual operational tasks (not just financial approvals) is uncommon in CRM products. It surfaces accountability exactly where errors occur.

Contextual AI assistant. The Groq integration is not a generic chatbot. It receives structured context about the user's active weddings, pending tasks, and leads on every query, making responses operationally useful rather than generically informative.

Template-driven onboarding. Pre-scaffolded wedding templates reduce time-to-value for new planners significantly, particularly for smaller firms that lack the capacity to build process structures from scratch.

Unified notification pipeline. In-app alerts, email reminders, and SMS messages are triggered by the same underlying data, consolidating three separate manual follow-up channels into one automated flow.

10. Scalability & Future Scope

10.1 Current Scalability Posture

The architecture is designed for growth from a single-firm deployment to a multi-tenant SaaS product.

- MongoDB horizontal scaling via Atlas sharding accommodates data growth without

schema changes.

- Stateless JWT authentication allows the API tier to scale horizontally behind a load balancer.
- Zustand's modular store architecture allows new feature modules to be added without refactoring existing state.
- The cron-based notification system can be migrated to a message queue (BullMQ, AWS SQS) as volume grows.

10.2 Planned Enhancements

- **Client portal.** A white-labeled client-facing view with real-time progress, mood board access, and document signing.
- **Multi-tenancy.** Organisation-level data isolation to support multiple planning firms on shared infrastructure.
- **Mobile application.** A React Native client reusing the existing API without backend modifications.
- **Calendar integration.** Bidirectional sync with Google Calendar and Outlook for events and task deadlines.
- **Financial reporting.** Exportable profit-and-loss summaries per wedding with payment receipt tracking.
- **AI vendor matching.** Using historical task completion data and ratings to recommend vendors for new weddings.
- **Offline support.** Progressive Web App capabilities with service workers for planners in venues with poor connectivity.

11. Conclusion

Aayojan demonstrates that a domain-specific CRM, built with precision around the operational reality of a single industry, delivers substantially more value than a horizontal tool adapted post-hoc. By combining a clean role-based data model, automated vendor communication, contextual AI assistance, and a carefully designed interface, the platform addresses the three core pain points of professional wedding planners: scattered communication, lack of progress visibility, and operational chaos under scale.

The system is fully functional, production-ready in architecture, and designed to grow from a single-firm tool to a multi-tenant platform. It represents a practical, technically grounded solution to a genuine and underserved market problem.